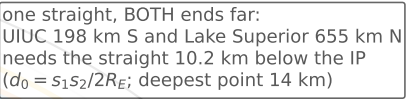


35 m-deep straight; one colour per tilt – dotted: the end that goes up, solid: the end that goes down; at 0 mrad they are the same curve



**baseline, up-going end:
2.2 km, inside the fence**

0.2

0



0 mrad (both ends)

5 mrad

2 mrad

ad

50 mrad

SW

**baseline, down-going end:
UIUC 198 km**

15.4 mrad

65 mrad

- up-going end surfaces here (near emergence) ■ Fermilab site — the rings in plan (collider, RCS3/4, RCS1/2)
 — down-going end surfaces here (far exit) ■ lakes (Natural Earth 10m) - - - a chord with both ends far (10 km deep)
 — same, smooth sphere ■ one straight: its two ends are opposite each other

Bold curves: GEBCO 2020 terrain along 144 bearings (0.25–1000 km), Catmull-Rom smoothed; exits over the Great Lakes taken at the water surface. Faint circles: smooth sphere, $s_{near} = R_E(-\theta + \sqrt{\theta^2 + 2d_0/R_E}) \approx d_0/\theta$ and $s_{far} = R_E(\theta + \sqrt{\theta^2 + 2d_0/R_E}) \approx 2R_E\theta$. Log range; rings in km. The fence is 0.9–4.9 km out, so the dotted curves show which tilts emerge on site.